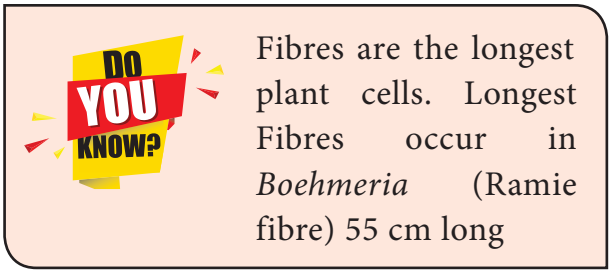
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b. Fibre tracheids c. Septate fibres d. Gelatinous fibres.



Fibres are the longest plant cells. Longest Fibres occur in *Boehmeria* (Ramie fibre) 55 cm long

a. **Libriform fibres:** These fibres have slightly lignified secondary walls with simple pits. These fibres are long and narrow.
b. **Fibre tracheids:** These are shorter than the libriform fibres with moderate secondary thickenings in the cell walls. Pits are simple or bordered.
c. **Septate fibres:** Fibres that have thin septa separating the lumen into distinct chambers. Eg. Teak
d. **Gelatinous fibres:** Fibres in which lignin is less in amount and cellulose is more in this cell walls.

These fibres are characteristic of tension wood which is formed in the underside of leaning stems and branches.

Bastfibres or Extra Xylary Fibres

These fibres are present in the phloem. Natural Bast fibres are strong and cellulosic. Fibres obtaining from the phloem or outer bark of jute, kenaf, flax and hemp plants. The so called pericyclic fibres are actually phloem fibres.

Surface Fibres

These fibres are produced from the surface of the plant organs. Cotton and silk cotton are the examples.They occur in the testa of seeds.

Mesocarp Fibres

Fibres obtained from the mesocarp of drupes like Coconut.

Leaf Fibres

Fibres obtained from the leaf of *Musa, Agave and Sensciveria.*

Fibres in Our Daily Life

Economically fibres may be grouped as follows

1. **Textile Fibres:** Fibres utilized for the manufacture of fabrics, netting and cordage etc.
   a. **Surface Fibres**: Example: Cotton.
   b. **Soft Fibres**: Example: Flax, Jute and Ramie
   c. **Hard fibres:** Example: Sisal, Coconut, Pineapple, Abaca *etc.*
2. **Brush fibre:** Fibres utilized for the manufacture of brushes and brooms.
3. **Rough weaving fibres:** Fibres utilized in making baskets, chairs, mats etc.
4. **Paper making fibres:** Wood fibres utilized for paper making.
5. **Filling fibres:** Fibres used for stuffing cushions, mattresses, pillows, furniture etc. Example: *Bombax* and Silk cotton.

Complex Tissues

A complex tissue is a tissue with several types of cells but all of them function together as a single unit. It is of two types – xylem and phloem.



Xylem

The xylem is the principal water conducting tissue in a vascular plant. The term xylem was introduced by **Nageli**(1858) and is